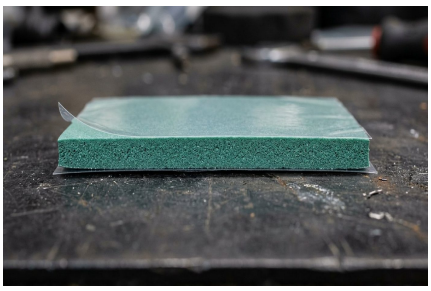
ULTRA SERIES (Ultra-550 / Ultra-NS Non-Silicone)

InsulTherm Ultra™

High-Flux Bimodal Al2O3 + Aligned h-BN Hybrid Percolation TIM Pad (5.5 W/m·K)

InsulTherm Ultra™ is an ultra-low thermal impedance hybrid percolation gap filler pad engineered for high-heat-flux 800V Silicon-Carbide (SiC) power modules and 100kW+ AI data center liquid cold plates where conventional 3.0 W/m·K alumina pads choke. Combining an 82% wt bimodal spherical alumina network (2 µm + 20 µm) with shear-aligned hexagonal Boron Nitride (h-BN) platelets inside a 35 Shore 00 ultra-wetout matrix, Ultra™ achieves R_th < 0.15 K·cm²/W at 20 psi clamping pressure.



Macro Cross-Section / Studio Hardware Inspection

k = 5.5 W/m·K	Rth < 0.15 K·cm²/W	35 Shore 00	Aligned h-BN	Non-Silicone Opt.

1. CHEMICAL / MATERIAL COMPOSITION & INTERNAL LAYER ARCHITECTURE

Ultra-550 utilizes a functionalized ultra-low-modulus siloxane copolymer loaded at 82% wt with bimodal spherical Al2O3 (interstitial 2 µm spheres bridging 20 µm primary spheres) plus aligned hexagonal Boron Nitride (h-BN, intrinsic k > 300 W/m·K) platelets forming continuous phonon highways. Ultra-NS substitutes a 100% siloxane-free acrylic elastomer for zero volatile migration on optics.

Layer 1 (Top Liner)	Diamond-Embossed Low-Tack PET Release Film (Micro-Air Channel Evacuation)
Layer 2 (Contact Skin)	Ultra-Wetout 35 Shore 00 Low-Modulus Interfacial Contact Skin
Layer 3 (Hybrid Core)	82% wt Bimodal Al2O3 (2µm+20µm) + Aligned h-BN Percolation Network (5.5 W/m·K)
Layer 4 (Carrier)	75 µm Antistatic Release Carrier Liner (Cleanroom Silicone-Free Option)

2. TYPICAL PHYSICAL, THERMAL & ELECTRICAL PROPERTIES

PROPERTY / PARAMETER	TYPICAL ENGINEERING VALUE	TEST METHOD / STANDARD
Bulk Thermal Conductivity	5.5 W/m·K	ASTM D5470-17 Steady-State
Thermal Impedance @ 20 psi	< 0.15 K·cm²/W (@ 0.5 – 1.0 mm nominal)	ASTM D5470-17 Calibration Rig
Hardness / Surface Wetout	35 Shore 00 (Ultra-conformable wetout)	ASTM D2240
Dielectric Breakdown Strength	> 6.5 – 12.0 kV/mm	ASTM D149
Volume Resistivity	> 1.0 × 10^13 Ω·cm	ASTM D257
Flammability Rating	UL 94 V-0	UL 94
Siloxane Volatility (D3–D10)	< 100 ppm (Ultra-550) | 0 ppm (Ultra-NS)	GC-MS Headspace
Specific Gravity	3.15 ± 0.10 g/cm³	ASTM D792
Standard Thicknesses	0.5 mm, 1.0 mm, 1.5 mm, 2.0 mm, 2.5 mm	ASTM D374
Operating Temperature Range	-55°C to +200°C (550) / -40°C to +125°C (NS)	IEC 60216

InsulTherm Innovations, Advanced Materials & Digital Converting Wing, Block C-14, Sanand GIDC Industrial Estate (Near iCreate Campus), Ahmedabad, Gujarat – 382110, India

Engineering Notice: Typical laboratory & baseline metrics (ASTM D5470 / UL 94 / ASTM E595). System-level OEM qualification is required prior to volume release.

3. ORDERING NOMENCLATURE, PREMIX VARIANTS & COMMERCIAL PRICING SCHEDULE

PART / GRADE CODE	MATERIAL SPECIFICATION & CONSTRUCTION	PROTOTYPE / R&D TIER	VOLUME / OEM TIER
`Ultra-550 (Area-Rated)`	82% Bimodal Al2O3 + Aligned h-BN (100 cm² ref. = ₹140; ₹1.40/cm² proto)	**₹140 / 100cm² (min ₹50/pc)**	**₹95 / 100cm² (₹0.95/cm² vol)**
`Ultra-NS (Non-Silicone)`	100% Siloxane-Free Optical/AI Grade (100 cm² ref. = ₹140; ₹1.40/cm² proto)	**₹140 / 100cm² (min ₹50/pc)**	**₹95 / 100cm² (₹0.95/cm² vol)**
`Inverter / SiC Kit`	Power Electronics, Rail & SiC Isolation Kit (TO-247 / EconoDUAL + Shield™)	**₹999 (25-pc) | ₹7,500 (Line)**	**Ships in 48 Hours**

4. TARGET INDUSTRIAL APPLICATIONS & DEPLOYMENT SECTORS

- 100kW+ AI data center liquid cold plates, high-TDP GPU/ASIC server accelerator blocks, and localized compute racks.
- 800V Silicon-Carbide (SiC) MOSFET & IGBT traction power modules (TO-247, EconoDUAL, PrimePACK) in EV and rail inverters.
- 800G / 1.6T coherent optical transceivers, telecom laser pump modules, and precision avionics optics (Ultra-NS siloxane-free grade).
- Medical imaging MRI gradient amplifiers and high-frequency electrosurgical generators.

5. DIGITAL CNC CONVERTING, APPLICATION & STORAGE GUIDELINES

1. Micro-Void Elimination: Apply Ultra™ using a rolling motion from one edge to expel trapped air via the diamond-embossed channels. 2. Clamping Pressure: Achieves optimal interfacial wetout (<0.15 K·cm²/W) at just 15–25 psi, protecting fragile bare-die SiC and BGA packages. 3. Non-Silicone Cleanliness: Specify 'Ultra-NS' on all optical/laser BOMs; processed on dedicated cleanroom silicone-free CNC vacuum mats.

6. QUALITY ASSURANCE, REGULATORY COMPLIANCE & SOVEREIGN DVA TRACEABILITY

Thermal & Electrical Metrology	Batch-wise ASTM D5470-17 thermal impedance calibration curves (10–50 psi) & ASTM D149 dielectric verification.
Flammability & Fire Safety	UL 94 V-0 rated material formulations; supports ARAI AIS-156 Amendment 3 & UL 9540A pack/rack propagation testing.
Outgassing & Environmental	RoHS 3 (EU 2015/863) & REACH compliant; ASTM E595 low-outgassing trace on Graphite-Flex™ & Ultra-NS™ grades.
Domestic Value Addition (DVA)	100% India Digital CNC Converting (Gujarat); compliant with PM E-DRIVE, SECI BESS DVA & <₹200 Cr MoD GTE Ban.

RAPID 24–48 HOUR CAD-TO-PART PROTOTYPING & OEM PROCUREMENT DESK
Submit .DXF / .STEP drawings, request <₹15,000 Petty-Cash R&D Sampler Kits, or schedule an engineering review:
Primary Official Desk: manav.sb@inslthrm.in | **Active Operations & Order Routing:** manavsb9719@gmail.com
Engineering Helpline: +91 78148 XXXXX | **Manufacturing Wing:** InsulTherm Innovations, Advanced Materials & Digital Converting Wing, Block C-14, Sanand GIDC Industrial Estate (Near iCreate Campus), Ahmedabad, Gujarat – 382110, India